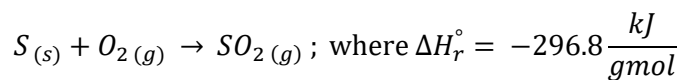


NOVEMBER 26, 2019

Liquid sulfur at 130°C and pure oxygen at 25°C are mixed at a stoichiometric ratio and fed to a burner, where oxidation of sulfur to SO₂ takes place. The burner is perfectly insulated, the pressure is 1 atm, and the system operates at steady state. What is the temperature of the exiting stream?

Useful information:



$$\Delta \hat{H}_m = 1.73 \frac{kJ}{gmol} \text{ for S at } 114^\circ\text{C}$$

$$C_{p,S(l)} = 32 \frac{J}{gmol \cdot ^\circ\text{C}} \quad C_{p,SO_2(g)} = 39.9 \frac{J}{gmol \cdot ^\circ\text{C}}$$

$$C_{p,S(s)} = 23.2 \frac{J}{gmol \cdot ^\circ\text{C}} \quad C_{p,O_{2(g)}} = 29.3 \frac{J}{gmol \cdot ^\circ\text{C}}$$